

Schur product of additive cyclic codes

Burcu Ecem Karakaş

joint work with Ferruh Özbudak and Buket Özkaya

The 14th International Workshop on Coding and Cryptography

Cryptography, Institute of Applied Mathematics

Middle East Technical University, Ankara

burcuey@metu.edu.tr

June 11, 2026

Overview

- 1 Schur Product
- 2 Cyclic Codes
- 3 Schur Product in Index 2
- 4 Schur Product in Index 3
- 5 Schur Product in Arbitrary Index
- 6 Dimension and Distance of Schur Product
- 7 References

Schur product

- Given two vectors $u = (u_0, \dots, u_{m-1})$, $v = (v_0, \dots, v_{m-1}) \in \mathbb{F}_q^m$, we define their *Schur product*, denoted by $u \star v$, as

$$u \star v = (u_0 \cdot v_0, \dots, u_{m-1} \cdot v_{m-1}).$$

- If $\mathcal{C}_1$ and $\mathcal{C}_2$ are two linear codes of the same length over the same field, then their Schur product is

$$\mathcal{C}_1 \star \mathcal{C}_2 := \text{Span}_{\mathbb{F}_q} \{c_1 \star c_2 : c_1 \in \mathcal{C}_1, c_2 \in \mathcal{C}_2\}.$$

- If G_1 and G_2 are generator matrices of $\mathcal{C}_1$ and $\mathcal{C}_2$ with k_1 and k_2 rows, respectively, then

$$\mathcal{C}_1 \star \mathcal{C}_2 = \text{Span}_{\mathbb{F}_q} \{r_i \star s_j : 1 \leq i \leq k_1, 1 \leq j \leq k_2\}$$

so that r_i and s_j denote the i^{th} and j^{th} rows of G_1 and G_2 , respectively.

Cyclic Codes

- An $[m, k]_q$ -linear code $\mathcal{C}$ is a cyclic code if $c = (c_0, c_1, \dots, c_{m-1}) \in \mathcal{C}$ implies $(c_{m-1}, c_0, c_1, \dots, c_{m-2}) \in \mathcal{C}$.
- Let $R := \mathbb{F}_q[x]/\langle x^m - 1 \rangle$ and $g(x) \in R$ be the generator polynomial of the $[m, k]_q$ -cyclic code $\mathcal{C}$ with $\deg(g(x)) = m - k$.
- $\mathcal{C} = \langle g(x) \rangle$ and $\mathcal{C}^\perp = \langle h^*(x) \rangle$, where $h(x) = \frac{x^m - 1}{g(x)}$ is a parity-check polynomial and $h^*(x) = x^k h\left(\frac{1}{x}\right)$.
- Let $m \geq 1$ such that $\gcd(m, q) = 1$ from this point on. Let r be the smallest integer satisfying $q^r \equiv 1 \pmod{m}$ and let $\mathbb{F}_{q^r}$ be the extension of $\mathbb{F}_q$ such that $\mathbb{F}_{q^r} = \mathbb{F}_q(\xi)$, where $\xi \in \mathbb{F}_{q^r}$ is a primitive m^{th} root of unity (i.e., $\mathbb{F}_{q^r}$ is the splitting field of $x^m - 1$ over $\mathbb{F}_q$).

Cyclic Codes as an Evaluation Codes

For $w, p \in \mathbb{Z}^+$, we now set some notation:

- zero set of $\mathcal{C}$: $Z(\mathcal{C}) = \{\xi^j \in \mathbb{F}_{q^r} : g(\xi^j) = 0\} = \{\xi^{j_1}, \dots, \xi^{j_w}\},$
- defining set of $\mathcal{C}$: $J = \{j_1, \dots, j_w\},$
- generating set of $\mathcal{C}$: $I = \Omega_m \setminus J$, where $\Omega_m = \{0, 1, \dots, m-1\},$
- zero set of $\mathcal{C}^\perp$: $Z(\mathcal{C}^\perp) = \{\xi^i \in \mathbb{F}_{q^r} : h^*(\xi^i) = 0\} = \{\xi^{i_1}, \dots, \xi^{i_p}\}.$

Theorem 1

Let $\mathcal{C}$ be an $[m, k]_q$ -cyclic code over $\mathbb{F}_q$ and ξ be the primitive m^{th} root of unity, $\xi \in \mathbb{F}_{q^r}$ and $\gcd(m, q) = 1$. Assume that the defining set of $\mathcal{C}$ is $J = \{j_1, \dots, j_w\}$ and the generating set of $\mathcal{C}$ is $I = \Omega_m \setminus J = \{i_1, \dots, i_t\}$. Then the defining set of $\mathcal{C}^\perp$ is $-I = \{-i_1, \dots, -i_t\} = \{m - i : i \in I\}.$

Cyclic Codes as an Evaluation Codes

Theorem 2

(Cascardo, 2018, Lemma 4) Let $\mathcal{C}$ be a $[m, k]$ -cyclic code over $\mathbb{F}_q$. Then $B(-I)|_{\mathbb{F}_q} = \mathcal{C}$ such that

$$\begin{aligned} B(-I) &= \left\{ (f(1), f(\xi), \dots, f(\xi^{m-1})) : f(x) \in \mathbb{F}_{q^r}[x], f(x) = \sum_{j \in -I} f_j x^j \right\} \\ &= \left\{ \left(\sum_{j \in -I} f_j \xi^{ij} \right)_{0 \leq i \leq m-1} : f_j \in \mathbb{F}_{q^r}, \forall j \right\}. \end{aligned}$$

In the polynomial representation over R , let $u(x) = \sum_{i \in \Omega} u_i x^i$ and $v(x) = \sum_{i \in \Omega} v_i x^i$, then $u(x) \star v(x) = \sum_{i \in \Omega} u_i v_i x^i$.

Theorem 3

(Cascardo, 2018, Lemma 6) $\mathcal{C} \star \mathcal{C}$ is a cyclic code with generator polynomial

$$g'(x) = \gcd\{g \star g, g \star (g \cdot x), g \star (g \cdot x^2), \dots, g \star (g \cdot x^{n-\deg(g)-1})\}.$$

Schur Product of cyclic codes

Theorem 4

(Casado, 2018, Theorem 1) Assume that, C_1 and C_2 are two cyclic codes of length m over $\mathbb{F}_q$ with generating sets l_1 and l_2 , respectively. Let

$$B(-l_1) = \left\{ \left(\sum_{j_1 \in -l_1} f_{j_1} \xi^{ij_1} \right)_{0 \leq i \leq m-1} : f_{j_1} \in \mathbb{F}_{q^r}, \forall j_1 \right\},$$

$$B(-l_2) = \left\{ \left(\sum_{j_2 \in -l_2} g_{j_2} \xi^{ij_2} \right)_{0 \leq i \leq m-1} : g_{j_2} \in \mathbb{F}_{q^r}, \forall j_2 \right\}.$$

Then we have $B(-(l_1 + l_2))|_{\mathbb{F}_q} = C_1 \star C_2$, where $l_1 + l_2 = \{i_1 + i_2 : i_1 \in l_1, i_2 \in l_2\}$.

Schur Product of cyclic codes

We now generalize (Cascudo, 2018, Definition 10) as follows.

- Given any $P \in \mathcal{P}(\Omega_m)$, where $\mathcal{P}(\Omega_m)$ be the power set of Ω_m , the amplitude of P will be denoted by $\text{amp}(P)$ and it is defined as

$$\text{amp}(P) := \min \{|E| : P \subseteq E \text{ and } E \text{ is consecutive}\}.$$

- $\text{amp}(P)$ is the size of the smallest consecutive set in Ω_m that contains P .
- If $\text{amp}(I) = |E|$ such that I is the generating set of the cyclic code $\mathcal{C}$, for some consecutive set $E \subseteq \Omega_m$, then $\Omega_m \setminus E$ is the largest consecutive set contained in its defining set J .

Theorem 5

(Cascudo, 2018, Theorem 2) Let $\mathcal{C} = B(-I)|_{\mathbb{F}_q}$ be a cyclic code with generating set I . Then

- $\dim_{\mathbb{F}_q}(\mathcal{C}) = |I|$ and $\dim_{\mathbb{F}_q}(\mathcal{C} \star \mathcal{C}) = |I + I|$.
- $d(\mathcal{C}) \geq m - \text{amp}(I) + 1$ and $d(\mathcal{C} \star \mathcal{C}) \geq m - \text{amp}(I + I) + 1$.

$\mathbb{F}_q$ -linear additive cyclic code over $\mathbb{F}_{q^2}$ of length n

- Assume that $\mathcal{C}$ is an $\mathbb{F}_q$ -linear additive cyclic code over $\mathbb{F}_{q^2}$ of length n ,
- Such an additive cyclic code can be viewed as a quasi-cyclic code of index 2 and of length n ,
- The polynomial generator matrix of an additive cyclic code of length n over $\mathbb{F}_{q^2}$ is of the form (Shi et al., [2022](#), Theorem 3.2),

$$\begin{bmatrix} p(x) & q(x) \\ 0 & r(x) \end{bmatrix}, \quad (1)$$

where $p(x), q(x), r(x) \in \mathbb{F}_q[x]/\langle x^n - 1 \rangle$ with $p(x)|(x^n - 1)$, $r(x)|(x^n - 1)$ and $\deg(q(x)) < \deg(r(x))$.

Schur Product of Index 2

Algorithm 1 Polynomial Generator Matrix Schur Product of Index 2

Input: $p(x), q(x), r(x)$

- 1: $d_p = n - \deg(p(x)) - 1$
 - 2: $d_r = n - \deg(r(x)) - 1$
 - 3: $p^{(1)}(x) \leftarrow \gcd(p(x) \star p(x), p(x) \star xp(x), \dots, p(x) \star x^{d_p} p(x))$
 - 4: $r^{(1)}(x) \leftarrow \gcd(r(x) \star r(x), r(x) \star xr(x), \dots, r(x) \star x^{d_r} r(x))$
 - 5: $a_0(x), \dots, a_{d_p}(x) \leftarrow \text{ExtendedEuclideanAlg}((p(x) \star p(x)), \dots, (p(x) \star x^{d_p} p(x)))$
 - 6: $q^{(1)}(x) \leftarrow a_0(x)(q(x) \star q(x)) + \dots + a_{d_p}(x)(q(x) \star x^{d_p} q(x))$
 - 7: **if** $d_p \geq d_r$ **then**
 - 8: $r^{(2)}(x) \leftarrow \gcd(r^{(1)}(x), q(x) \star r(x), xq(x) \star r(x), \dots, x^{d_p} q(x) \star r(x))$
 - 9: **else if** $d_p < d_r$ **then**
 - 10: $r^{(2)}(x) \leftarrow \gcd(r^{(1)}(x), q(x) \star r(x), q(x) \star xr(x), \dots, q(x) \star x^{d_r} r(x))$
 - 11: **end if**
 - 12: $(s_0(x), q^{(2)}(x)) \leftarrow \text{DivisionAlg}(q^{(1)}(x), r^{(2)}(x))$
 - 13: **return** $p^{(1)}(x), q^{(2)}(x), r^{(2)}(x)$
-

Schur Product of Index 2

Theorem 6

Let $\mathcal{C}$ be an $\mathbb{F}_q$ -linear additive cyclic code over $\mathbb{F}_{q^2}$ of length n with the polynomial generator matrix,

$$\begin{bmatrix} p(x) & q(x) \\ 0 & r(x) \end{bmatrix}.$$

Then the Schur product $\mathcal{C} \star \mathcal{C}$ is also additive cyclic with the polynomial generator matrix

$$\begin{bmatrix} p^{(1)}(x) & q^{(2)}(x) \\ 0 & r^{(2)}(x) \end{bmatrix},$$

where $p^{(1)}(x), q^{(2)}(x), r^{(2)}(x)$ can be found by using Algorithm 1.

Example

- Let the generator matrix of $\mathcal{C}$ be $\begin{bmatrix} p(x) & q(x) \\ 0 & r(x) \end{bmatrix} = \begin{bmatrix} x^2 + 1 & 1 \\ 0 & x - 1 \end{bmatrix}$.

- $d_p = n - \deg(p) - 1 = 4 - 2 - 1 = 1$

- $d_r = n - \deg(r) - 1 = 4 - 1 - 1 = 2$

$$\begin{bmatrix} p(x) & q(x) \\ x^{d_p} p(x) = xp(x) & xq(x) \\ 0 & r(x) \\ 0 & xr(x) \\ 0 & x^{d_r} r(x) = x^2 r(x) \end{bmatrix} = \begin{bmatrix} x^2 + 1 & 1 \\ x(x^2 + 1) & x(1) \\ 0 & (x - 1) \\ 0 & x(x - 1) \\ 0 & x^2(x - 1) \end{bmatrix} = \begin{bmatrix} x^2 + 1 & 1 \\ x^3 + x & x \\ 0 & x - 1 \\ 0 & x^2 - x \\ 0 & x^3 - x^2 \end{bmatrix},$$

- We get

$$\begin{aligned} p^{(1)}(x) &= \gcd(p(x) \star p(x), p(x) \star xp(x)) = \gcd(x^2 + 1, 0) \\ &= x^2 + 1 \\ &= 1(x^2 + 1) + 0(0). \end{aligned}$$

- Since p and q are in the same row,

$$\begin{aligned} q^{(1)}(x) &= 1(q(x) \star q(x)) + 0(q(x) \star xq(x)) \\ &= 1(1) + 0(0) \\ &= 1. \end{aligned}$$

Example

- Then,

$$\begin{aligned} r^{(1)}(x) &= \gcd(r(x) \star r(x), r(x) \star xr(x), r(x) \star x^2r(x)) \\ &= \gcd(x+1, -x, 0) \\ &= 1. \end{aligned}$$

- Since $d_p = 1 < d_r = 2$,

$$\begin{aligned} r^{(2)}(x) &= \gcd(r^{(1)}(x), q(x) \star r(x), q(x) \star xr(x), q(x) \star x^2r(x)), \\ &= \gcd(1, -1, 0, 0) \\ &= 1. \end{aligned}$$

- Next,

$$\begin{aligned} q^{(2)}(x) &= q^{(1)}(x) - s_0(x)r^{(2)}(x) \\ &= 1 - s_0(x)(1) \\ &= 1 - 1(1) \\ &= 0, \end{aligned}$$

such that $\deg(q^{(2)}(x)) < \deg(r^{(2)}(x))$. Therefore

$$\begin{aligned} &\begin{bmatrix} p(x) & q(x) \\ 0 & r(x) \end{bmatrix} \star \begin{bmatrix} p(x) & q(x) \\ 0 & r(x) \end{bmatrix} = \begin{bmatrix} p^{(1)}(x) & q^{(2)}(x) \\ 0 & r^{(2)}(x) \end{bmatrix} \\ \Rightarrow &\begin{bmatrix} x^2+1 & 1 \\ 0 & x-1 \end{bmatrix} \star \begin{bmatrix} x^2+1 & 1 \\ 0 & x-1 \end{bmatrix} = \begin{bmatrix} x^2+1 & 0 \\ 0 & 1 \end{bmatrix}. \end{aligned}$$

$\mathbb{F}_q$ -linear additive cyclic code over $\mathbb{F}_{q^3}$ of length n

- Assume that $\mathcal{C}$ is an $\mathbb{F}_q$ -linear additive cyclic code over $\mathbb{F}_{q^3}$ of length n ,
- Such an additive cyclic code can be viewed as a quasi-cyclic code of index 3 and of length n ,
- The polynomial generator matrix of an additive cyclic code of length n over $\mathbb{F}_{q^3}$ is of the form (see Shi, Chu, and Özbudak, [2025](#), Theorem 3.2),

$$\begin{bmatrix} p(x) & q(x) & m(x) \\ 0 & r(x) & n(x) \\ 0 & 0 & l(x) \end{bmatrix}, \quad (2)$$

where $p(x), q(x), r(x), m(x), n(x), l(x) \in \mathbb{F}_q[x]/\langle x^n - 1 \rangle$ with $p(x)|(x^n - 1)$, $r(x)|(x^n - 1)$, $l(x)|(x^n - 1)$, $\deg(q(x)) < \deg(r(x))$, $\deg(m(x)) < \deg(l(x))$ and $\deg(n(x)) < \deg(l(x))$.

Schur Product of Index 3

Algorithm 2 Polynomial Generator Matrix Schur Product of Index 3

Input: $p(x), q(x), m(x), r(x), n(x), l(x)$

```

1:  $d_p = n - \deg(p(x)) - 1$ 
2:  $d_r = n - \deg(r(x)) - 1$ 
3:  $d_l = n - \deg(l(x)) - 1$ 
4:  $p^{(1)}(x) \leftarrow \gcd(p(x) \star p(x), p(x) \star xp(x), \dots, p(x) \star x^{d_p}p(x))$ 
5:  $r^{(1)}(x) \leftarrow \gcd(r(x) \star r(x), r(x) \star xr(x), \dots, r(x) \star x^{d_r}r(x))$ 
6:  $l^{(1)}(x) \leftarrow \gcd(l(x) \star l(x), l(x) \star xl(x), \dots, l(x) \star x^{d_l}l(x))$ 
7:  $a_0(x), \dots, a_{d_p}(x) \leftarrow \text{ExtendedEuclideanAlg}((p(x) \star p(x)), \dots, (p(x) \star x^{d_p}p(x)))$ 
8:  $q^{(1)}(x) \leftarrow a_0(x)(q(x) \star q(x)) + \dots + a_{d_p}(x)(q(x) \star x^{d_p}q(x))$ 
9:  $m^{(1)}(x) \leftarrow a_0(x)(m(x) \star m(x)) + \dots + a_{d_p}(x)(m(x) \star x^{d_p}m(x))$ 
10:  $b_0(x), \dots, b_{d_r}(x) \leftarrow \text{ExtendedEuclideanAlg}((r(x) \star r(x)), \dots, (r(x) \star x^{d_r}r(x)))$ 
11:  $n^{(1)}(x) \leftarrow b_0(x)(n(x) \star n(x)) + \dots + b_{d_r}(x)(n(x) \star x^{d_r}n(x))$ 
12: if  $d_r \geq d_l$  then
13:    $l^{(2)}(x) \leftarrow \gcd(l^{(1)}(x), n(x) \star l(x), xn(x) \star l(x), \dots, x^{d_r}n(x) \star l(x))$ 
14:    $n^{(2)}(x) \leftarrow \gcd(n^{(1)}(x), m(x) \star n(x), xm(x) \star n(x), \dots, x^{d_p}m(x) \star n(x))$ 

```

```

15: else if  $d_r < d_l$  then
16:    $l^{(2)}(x) \leftarrow \gcd(l^{(1)}(x), n(x) \star l(x), n(x) \star xl(x), \dots, n(x) \star x^{d_l}l(x))$ 
17:    $n^{(2)}(x) \leftarrow \gcd(n^{(1)}(x), m(x) \star n(x), m(x) \star xn(x), \dots, m(x) \star x^{d_r}n(x))$ 
18: end if
19: if  $d_p \geq d_r$  then
20:    $r^{(2)}(x) \leftarrow \gcd(r^{(1)}(x), q(x) \star r(x), xq(x) \star r(x), \dots, x^{d_p}q(x) \star r(x))$ 
21: else if  $d_p < d_r$  then
22:    $r^{(2)}(x) \leftarrow \gcd(r^{(1)}(x), q(x) \star r(x), q(x) \star xr(x), \dots, q(x) \star x^{d_r}r(x))$ 
23: end if
24:  $(s_1(x), q^{(2)}(x)) \leftarrow \text{DivisionAlg}(q^{(1)}(x), r^{(2)}(x))$ 
25:  $m^{(2)}(x) \leftarrow m^{(1)}(x) - s_1(x)n^{(2)}(x)$ 
26: if  $d_p \geq d_l$  then
27:    $l^{(3)}(x) \leftarrow \gcd(l^{(2)}(x), m(x) \star l(x), xm(x) \star l(x), \dots, x^{d_p}m(x) \star l(x))$ 
28: else if  $d_p < d_l$  then
29:    $l^{(3)}(x) \leftarrow \gcd(l^{(2)}(x), m(x) \star l(x), m(x) \star xl(x), \dots, m(x) \star x^{d_l}l(x))$ 
30: end if
31:  $(s_2(x), n^{(3)}(x)) \leftarrow \text{DivisionAlg}(n^{(2)}(x), l^{(3)}(x))$ 
32:  $(s_3(x), m^{(3)}(x)) \leftarrow \text{DivisionAlg}(m^{(2)}(x), l^{(3)}(x))$ 
33: return  $p^{(1)}(x), q^{(2)}(x), r^{(2)}(x), m^{(3)}(x), n^{(3)}(x), l^{(3)}(x)$ 

```

Schur Product of Index 3

Theorem 7

Let $\mathcal{C}$ be an $\mathbb{F}_q$ -linear additive cyclic code over $\mathbb{F}_{q^3}$ of length n with the polynomial generator matrix,

$$\begin{bmatrix} p(x) & q(x) & m(x) \\ 0 & r(x) & n(x) \\ 0 & 0 & l(x) \end{bmatrix}.$$

Then the Schur product $\mathcal{C} \star \mathcal{C}$ is also additive cyclic with the polynomial generator matrix

$$\begin{bmatrix} p^{(1)}(x) & q^{(2)}(x) & m^{(3)}(x) \\ 0 & r^{(2)}(x) & n^{(3)}(x) \\ 0 & 0 & l^{(3)}(x) \end{bmatrix},$$

where $p^{(1)}(x), q^{(2)}(x), r^{(2)}(x), m^{(3)}(x), n^{(3)}(x), l^{(3)}(x)$ can be found by using Algorithm 2.

Additive Cyclic Codes of Arbitrary Index

- Let $\mathbb{F}_{q^r}$ be the extension of the finite field $\mathbb{F}_q$, where r is a positive integer.
- Let $n|q^r - 1$ and $A = \{i_1, \dots, i_s\} \subseteq \Omega_n = \mathbb{Z}/n\mathbb{Z}$.
- Let α be a generator of subgroup of the order n of $\mathbb{F}_{q^r}$, say W .
- Define the space of polynomials $P(A)$ as

$$P(A) := \left\{ a_1 x^{i_1} + \dots + a_s x^{i_s} : a_1, \dots, a_s \in \mathbb{F}_{q^r} \right\}$$

- and

$$B(A) := \left\{ (f(\alpha^0), \dots, f(\alpha^{n-1})) : f(x) \in P(A) \right\} \subset \mathbb{F}_{q^r}^n.$$

- Let $m \leq r$ be a positive integer and $\tau = \{\gamma_1, \dots, \gamma_m\}$ be a linearly independent subset of $\mathbb{F}_{q^r}$ over $\mathbb{F}_q$.
- ϕ_τ be a surjective $\mathbb{F}_q$ -linear map from $\mathbb{F}_{q^r}$ to $\mathbb{F}_q^m$ such that $\phi_\tau(x) := (\text{Tr}(\gamma_1 x), \dots, \text{Tr}(\gamma_m x))$ where Tr is a trace map from $\mathbb{F}_{q^r}$ to $\mathbb{F}_q$.

Schur Product of Arbitrary Index

Then,

$$\begin{aligned}\phi_\tau(B(A)) &= \left\{ \phi_\tau(f(\alpha^0), \dots, f(\alpha^{n-1})) : f(x) \in P(A) \right\} \subset \mathbb{F}_q^{mn} \\ &= \left\{ \left(\text{Tr}(\gamma_1 f(\alpha^0)), \dots, \text{Tr}(\gamma_m f(\alpha^0)), \dots, \right. \right. \\ &\quad \left. \left. \text{Tr}(\gamma_1 f(\alpha^{n-1})), \dots, \text{Tr}(\gamma_m f(\alpha^{n-1})) \right) : f(x) \in P(A) \right\},\end{aligned}\quad (3)$$

is an additive cyclic code of length n over $\mathbb{F}_q^m$ (Güneri, Özbudak, and Özdemiir, [2017](#)).

Hasse-Weil Theorem for Additive Cyclic Codes

Theorem 8

(Güneri, Özbudak, and Özdemir, 2017, Theorem 1) Consider the additive cyclic code $C = \phi_\tau(B(A))$ of length $n = q^r - 1$ over E , where $A = \{i_1, \dots, i_s\} \subseteq \mathbb{Z}/n\mathbb{Z}$. Assume that $\gcd(i_j, q) = 1$ for all j and let $i = \max \{i_j : 1 \leq j \leq s\}$. Then,

$$d(C) \geq q^r - q^{r-m} - \frac{(q^m - 1)(i - 1)\lfloor 2\sqrt{q^r} \rfloor}{2q^m}.$$

When $n \mid (q^r - 1)$,

$$d(C) \geq \frac{n}{q^r - 1} \left(q^r - q^{r-m} - \frac{(q^m - 1)(i - 1)\lfloor 2\sqrt{q^r} \rfloor}{2q^m} \right),$$

where $i = \max \left\{ \frac{q^r - 1}{n} i_j \pmod{q^r - 1} : 1 \leq j \leq s \right\}$.

Schur Product of Arbitrary Index

We have found that

$$\phi_\tau(B(A)) \star \phi_\tau(B(A)) \subseteq \phi_\tau(B(A + \bar{A})),$$

where

$$A + \bar{A} = \{i_{j_1} + q^\ell i_{j_2} : 0 \leq \ell \leq r', 1 \leq j_1, j_2 \leq s\} \subseteq \bar{A} + \bar{A} \subseteq \tilde{A} + \tilde{A},$$

$\bar{A} := \{q^\ell i_j : 0 \leq \ell \leq r', 1 \leq j \leq s\}$ and $\tilde{A} := \{q^\ell i_j : 0 \leq \ell \leq r-1, 1 \leq j \leq s\}$ is the Galois closure of A .

Theorem 9

Let $\phi_\tau(B(A))$ be an additive cyclic code of length n over $\mathbb{F}_q^m$, then the Schur product $\phi_\tau(B(A)) \star \phi_\tau(B(A))$ is again additive cyclic and it is defined as

$$\phi_\tau(B(A)) \star \phi_\tau(B(A)) = \{\phi_\tau(h(\alpha^0), \dots, h(\alpha^{n-1})) : h(x) \in P(A + \bar{A})\}$$

such that $\phi_\tau(B(A)) \star \phi_\tau(B(A)) \subset \phi_\tau(B(A + \bar{A}))$ with

$$h(x) = \gamma_i^1 f(x)g(x) + \gamma_i^q (f(x)g(x)^q + f(x)^q g(x)) + \dots + \gamma_i^{q^{r'}} (f(x)g(x)^{q^{r'}} + f(x)^{q^{r'}} g(x)),$$

where $f(x), g(x) \in P(A)$, and

$$r' = \begin{cases} \frac{r-1}{2}, & \text{if } r \text{ is odd,} \\ \frac{r}{2}, & \text{if } r \text{ is even.} \end{cases}$$

Dimension and Distance of Schur Product

Theorem 10

Let $\mathcal{C} = \phi_\tau(B(A))$ be an $\mathbb{F}_q$ -linear additive cyclic code of length n over $\mathbb{F}_q^m$. Let E and $\bar{E}$ be the largest consecutive sets contained in A and $\bar{A}$, respectively. Then

- ① $\dim_{\mathbb{F}_q}(\mathcal{C}) \geq |E|$ and $\dim_{\mathbb{F}_q}(\mathcal{C} \star \mathcal{C}) \geq |E| + |\bar{E}|$.
- ② $d(\mathcal{C}^\perp) \geq n - \text{amp}(\Omega_n \setminus A) + 1$ and $d((\mathcal{C} \star \mathcal{C})^\perp) \geq n - \text{amp}(\Omega_n \setminus (A + \bar{A})) + 1$.

Hasse-Weil Bound for Schur Product of Additive Cyclic Codes

Theorem 11

Let $n = q^r - 1$ and $A = \{i_1, \dots, i_s\} \subseteq \Omega_n$. Assume that $\gcd(i_j, q) = 1$ for all j , then $\gcd(i_{j_1} + q^\ell i_{j_2}, q) = 1, \forall j_1, j_2 \in \{1, \dots, s\}$ and $0 \leq \ell \leq r'$. Let i be the maximum of $A + \bar{A} = \{i_{j_1} + q^\ell i_{j_2} : 0 \leq \ell \leq r', 1 \leq j_1, j_2 \leq s\}$, then

$$d(C \star C) \geq q^r - q^{r-m} - \frac{(q^m - 1)(i - 1)\lfloor 2\sqrt{q^r} \rfloor}{2q^m}.$$

If $n \mid (q^r - 1)$, then we obtain

$$d(C \star C) \geq \frac{n}{q^r - 1} \left(q^r - q^{r-m} - \frac{(q^m - 1)(i - 1)\lfloor 2\sqrt{q^r} \rfloor}{2q^m} \right),$$

where $i = \max \left\{ \frac{q^r - 1}{n} (i_{j_1} + q^\ell i_{j_2}) \pmod{q^r - 1} : 0 \leq \ell \leq r', 1 \leq j_1, j_2 \leq s \right\}$.

References I



Cascudo, Ignacio (2018). “On squares of cyclic codes”. In: *IEEE Transactions on Information Theory* 65.2, pp. 1034–1047.



Güneri, Cem, Ferruh Özbudak, and Funda Özdemir (2017). “Hasse–Weil bound for additive cyclic codes”. In: *Designs, Codes and Cryptography* 82.1, pp. 249–263.



Shi, Minjia, Shixiao Chu, and Ferruh Özbudak (2025). “Additive cyclic codes over $\mathbb{F}_{q^3}$ ”. In: *Journal of Algebra and its Applications* 24.09, p. 2550339.



Shi, Minjia et al. (2022). “Additive cyclic complementary dual codes over $\mathbb{F}_4$ ”. In: *Finite Fields and Their Applications* 83, p. 102087.

Thank you for listening.

Any question?